Notes: Kogan (JFE, 2004)
Asset Prices and Real Investment

Contents

1 Big picture

- **Research question.** How do real investment frictions shape the time-series behavior of stock returns? More specifically, can investment irreversibility and adjustment costs explain why market-to-book ratios and investment rates predict conditional volatility and, potentially, expected returns?
- **Main idea.** Stock prices are not determined only by discount rates and cash-flow news in an exchange economy. In a production economy, the supply of capital can adjust endogenously, and the elasticity of that supply depends on real investment constraints.
- **Core mechanism.** When firms cannot disinvest, capital supply is inelastic in low-q states. Demand shocks must then be absorbed by prices, making stock returns more volatile and more exposed to systematic shocks. When investment can respond elastically, prices absorb less of the shock and volatility is lower.
- **Full-model twist.** Once the investment rate is bounded above, very high-q states also have inelastic capital supply: firms want to expand, but cannot expand fast enough. Thus the model predicts a nonmonotonic, roughly U-shaped relation between market-to-book and conditional return volatility.
- **Empirical object.** The paper studies 13 U.S. industry portfolios and relates monthly return volatility proxies to lagged industry market-to-book ratios.
- **Main empirical finding.** The conditional volatility evidence supports the model: volatility is high when market-to-book is far below or far above its normal level, and lower near the middle. The evidence for expected returns is weak and statistically insignificant.
- **Economic takeaway.** Firm characteristics such as market-to-book are informative because they summarize the state of the real side of the economy. They reveal whether investment constraints are binding and therefore whether stock prices must absorb real shocks.

2 Incremental contribution of the paper

- **From exchange economies to production economies.** The paper moves asset-pricing intuition away from fixed-supply exchange-economy models and toward a production economy where capital supply is endogenous.
- **A real mechanism for return predictability.** Market-to-book is not just mechanically related to expected returns through price in the denominator or numerator. In the model, market-to-book is informative because it is a sufficient statistic for the investment state.
- **Irreversibility as an asset-pricing force.** The paper shows that irreversible investment can generate a leverage-effect-like relation between low prices and high volatility even when firms are financed entirely by equity.
- **Nonlinear prediction.** A central contribution is the prediction that the sign of the relation between market-to-book and conditional moments can change across states. This is different from standard linear empirical specifications.

- **Separation of irreversibility and bounded expansion.** Low-q states are governed by the inability to disinvest. High-q states are governed by the inability to invest fast enough. Both constraints create inelastic supply, but on opposite sides of the state space.
- **Empirical link to industry portfolios.** The paper takes the model to industry-level data, where investment irreversibility and adjustment frictions are more naturally interpreted than at the aggregate market level.
- **Volatility emphasis.** The paper highlights that the strongest predictions are for conditional volatility. Expected-return tests are secondary because expected returns are harder to estimate and depend on more stylized assumptions.

3 Model, data, and measurement

3.1 Two-sector production economy

The model is a two-sector production economy. The first sector represents the broad economy and has a stochastic constant-returns technology. The second sector represents an industry subject to investment frictions.

- **First sector.** The first good can be consumed, reinvested in the first sector, or transformed into second-sector capital.
- **Second sector.** The second sector uses its own capital to produce a sector-specific consumption good.
- **Irreversibility.** Once capital has been transferred into the second sector, it cannot be moved back to the first sector.
- **Equity financing.** Firms in the second sector are financed entirely by equity, so volatility effects are not driven by financial leverage.
- **Representative households.** Households choose consumption and portfolios over the first-sector technology, the second-sector stock, and a locally risk-free asset.

The basic capital-accumulation equations are

$$dK_{1,t} = (aK_{1,t} - c_{1,t})dt + \sigma K_{1,t} dW_t - dI_t,$$

$$dK_{2,t} = -\delta K_{2,t} dt + dI_t,$$

with irreversible investment $dI_t \geq 0$.

Interpretation. The model is designed to isolate one industry whose investment is hard to reverse. This lets Kogan study how that industry's stock price behaves when capital cannot freely move in and out.

3.2 State variable and Tobin's q

The economy can be summarized by the relative capital stock of the constrained sector:

$$X_t = b^{-1/\gamma} \frac{K_{2,t}}{K_{1,t}}, \qquad x_t = \log X_t.$$

Investment occurs when the second-sector capital stock is sufficiently low relative to the first-sector capital stock. Equivalently, investment occurs when Tobin's q reaches the investment threshold.

- **Average and marginal q.** In the model, marginal q equals average q, so the market value of a firm is $P_t = q_t K_{2,t}$.
- **Basic-model threshold.** With irreversible investment and instantaneous upward adjustment, $q_t \leq 1$ and investment occurs only when $q_t = 1$.
- **Market-to-book proxy.** Empirically, Tobin's q is proxied by the market-to-book ratio.

Interpretation. Market-to-book is informative because it tells us whether the industry has too much capital, is near the investment boundary, or is constrained in expanding.

3.3 Return decomposition

For a representative second-sector firm, the market value is $P_t = q_t K_{2,t}$. The cumulative return can be written as a profit component plus the change in q:

$$\frac{\pi_t dt + q_t dK_{2,t} + K_{2,t} dq_t - dI_t}{P_t} = \left(\frac{\pi_t}{P_t} - \delta\right) dt + \frac{dq_t}{q_t},$$

where investment occurs only when $q_t = 1$ in the basic irreversible-investment model.

- The profit yield reflects output-market conditions.
- The key volatility term is dq_t/q_t.
- When capital supply is inelastic, q_t must move more in response to shocks.

Interpretation. Return volatility is high when prices must do the adjustment that quantities cannot do.

3.4 Basic irreversibility mechanism

In the basic model, there are two economically important states.

- **Low q.** The industry has too much installed capital relative to current demand. Firms cannot disinvest, so capital supply is inelastic. Prices absorb shocks, and returns are volatile.
- **q near one.** Firms are close to investing. Positive demand shocks can be absorbed partly by new capital supply, so q is less sensitive to shocks and volatility is lower.

The model therefore predicts a negative relation between q and conditional volatility for $q \leq 1$.

3.5 Bounded investment rate

The full model adds an upper bound on the investment rate:

$$dK_{1,t} = (aK_{1,t} - c_{1,t})dt + \sigma K_{1,t} dW_t - i_t dt,$$

$$dK_{2,t} = -\delta K_{2,t} dt + i_t dt,$$

with

$$i_t \in [0, i_{\max} K_{2,t}].$$

Under the optimal policy, firms invest at the maximum rate when the industry capital stock is below the investment threshold, and invest zero otherwise.

- When $q < 1$, irreversibility binds and firms do not invest.
- Around $q \approx 1$, capital supply is relatively elastic.
- When $q > 1$, the industry wants to expand, but the investment-rate constraint binds.

Interpretation. With a bounded investment rate, q can exceed one. High q now indicates constrained expansion, so volatility rises again. This produces the paper's central nonmonotonic prediction.

3.6 Empirical data

The empirical analysis uses industry portfolios formed monthly from May 1963 through December 2001, giving 464 monthly observations.

- **Portfolios.** The sample consists of 13 industry portfolios formed from NYSE, Amex, and Nasdaq stocks.
- **Industry classification.** Stocks are sorted into industries using two-digit SIC codes.
- **Returns.** The tests use monthly excess returns, subtracting the one-month Treasury bill rate.
- **Market-to-book.** Industry market-to-book is computed as the inverse of the value-weighted book-to-market ratio for firms with Compustat data.
- **Main regressor.** The regressions use the natural logarithm of market-to-book, measured as a deviation from its industry mean.

Interpretation. Industry portfolios are a natural empirical level because investment frictions and capital specificity are often industry-level phenomena.

4 Methodology and empirical specifications

4.1 Model prediction for conditional volatility

The basic model implies that conditional return volatility is related to the slope of q with respect to the state variable:

$$\sigma_R = \sigma \left| \frac{q'(x)}{q(x)} \right|.$$

In the basic irreversible-investment model, this relation is monotone decreasing in q: low q means high volatility, and q near one means low volatility.

Economic logic. When firms are far from investing and cannot disinvest, quantities cannot respond to shocks. Prices respond more, so volatility is high.

4.2 Full-model prediction for nonmonotonicity

With a bounded investment rate, the model predicts three regions.

- **Low-q region.** Firms do not invest, and irreversibility prevents disinvestment. Supply is inelastic and volatility is high.
- **Intermediate-q region.** Firms are near the investment boundary. Capital supply is relatively elastic and volatility is low.
- **High-q region.** Firms are expanding at the maximum feasible rate. Supply is again inelastic and volatility is high.

Therefore, the relation between market-to-book and conditional volatility should be negative at low market-to-book and positive at high market-to-book.

4.3 Quadratic conditional-volatility regression

The first empirical specification is

$$|R_{i,t} - \bar{R}_i| = a_{i,0} + a_{i,1}(M/B)_{i,t-1} + a_{i,2}(M/B)^2_{i,t-1} + \varepsilon_{i,t}.$$

Here $|R_{i,t} - \bar{R}_i|$ proxies for conditional volatility of industry i at monthly frequency.

- $R_{i,t}$ is the excess monthly industry return.
- $\bar{R}_i$ is the sample mean industry return.
- $(M/B)_{i,t-1}$ is lagged log market-to-book, demeaned by industry.
- The model predicts $a_{i,2} > 0$.

Interpretation. A positive quadratic coefficient means volatility is higher when market-to-book is far from its normal level in either direction.

4.4 Piece-wise linear conditional-volatility regression

The second empirical specification is

$$|R_{i,t} - \bar{R}_i| = a_{i,0} + a_{i,1}(M/B)^-_{i,t-1} + a_{i,2}(M/B)^+_{i,t-1} + \varepsilon_{i,t},$$

where $(M/B)^-$ and $(M/B)^+$ denote the negative and positive parts of demeaned log market-to-book.

- The model predicts $a_{i,1} < 0$: when market-to-book is below normal, raising market-to-book lowers volatility.
- The model predicts $a_{i,2} > 0$: when market-to-book is above normal, raising market-to-book increases volatility.
- This specification tests the sign change directly.

Interpretation. This is the cleanest empirical test of the model's nonlinear state dependence, but it assumes the sign change occurs near the mean market-to-book ratio.

4.5 Conditional expected-return regression

The expected-return specification is

$$R_{i,t} = a_{i,0} + a_{i,1}(M/B)_{i,t-1} + a_{i,2}(M/B)^2_{i,t-1} + \varepsilon_{i,t}.$$

The model predicts $a_{i,2} > 0$, but this prediction is less direct than the volatility prediction.

- In the model, expected returns are linked to systematic risk.
- Systematic risk is related to conditional volatility.
- Empirically, expected returns are difficult to estimate in short samples.

Interpretation. The paper treats expected-return tests as secondary. Failure to find expected-return predictability does not overturn the central volatility mechanism.

4.6 Inference

The regressions are estimated separately for each of the 13 industry portfolios. The paper reports White standard errors and joint chi-square tests for whether the relevant coefficients are jointly equal to zero across industries.

5 Identification and interpretation

5.1 What the paper identifies

The paper identifies whether industry market-to-book ratios predict conditional volatility in the nonlinear way implied by a production economy with irreversible investment and adjustment costs. The empirical tests are not a randomized experiment; they are a theory-guided time-series test of model-implied state dependence.

5.2 Why market-to-book is informative

In the model, Tobin's q is a sufficient statistic for the constrained industry's state and investment policy. Empirically, market-to-book is used as a proxy for q. A low market-to-book ratio indicates that installed capital is worth less than replacement cost, so irreversibility is likely to matter. A high market-to-book ratio indicates strong demand for expansion, so the upper bound on investment may matter.

5.3 Why volatility results are central

The volatility implications follow directly from investment constraints and market clearing. If quantities cannot adjust, prices must adjust. Expected-return implications require additional assumptions about systematic risk, aggregate consumption, and the market price of risk. Kogan therefore emphasizes volatility as the sharper prediction.

5.4 Volatility versus exogenous uncertainty

The paper gives a different interpretation of evidence linking investment and stock-return volatility. Return volatility is not simply an exogenous proxy for uncertainty faced by firms. In the model, both investment and stock-return volatility are endogenous outcomes driven by the same state variable, q.

5.5 Interpretation of the coefficients

The quadratic coefficient $a_{i,2}$ in the volatility regression measures convexity in the relation between volatility and market-to-book. The piece-wise coefficients test the sign of the slope on each side of the mean market-to-book ratio. A negative low-side slope and positive high-side slope are consistent with the model's U-shaped volatility relation.

5.6 Limitations

- Market-to-book is an imperfect empirical proxy for Tobin's q.
- The volatility proxy $|R_{i,t} - \bar{R}_i|$ is simple and noisy.
- The piece-wise specification assumes the turning point occurs near the mean of market-to-book.
- The industry-level empirical design may miss firm-level heterogeneity.
- Expected-return tests have low power and do not support the model's expected-return prediction.

6 Tables and figures

6.1 Figures 1 and 2: Basic irreversible-investment model

Read Figure 1:

- Figure 1 plots conditional stock-return volatility against Tobin's q in the basic model.
- The relation is negative: volatility is highest at low q and falls as q approaches one.
- The three curves correspond to different preference parameter values.
- The figure resembles a leverage effect, but the mechanism is real investment irreversibility rather than financial leverage.

Read Figure 2:

- Figure 2 plots the conditional investment-to-capital ratio against Tobin's q in the basic model.
- Investment is essentially zero when q is far below one.
- Investment rises sharply as q approaches one.
- The figure shows why volatility falls near $q = 1$: capital supply becomes more elastic when firms are close to investing.

Economic message: In the basic model, low q means binding irreversibility, inelastic supply, and high return volatility. Near $q = 1$, investment can absorb shocks, so volatility is lower.

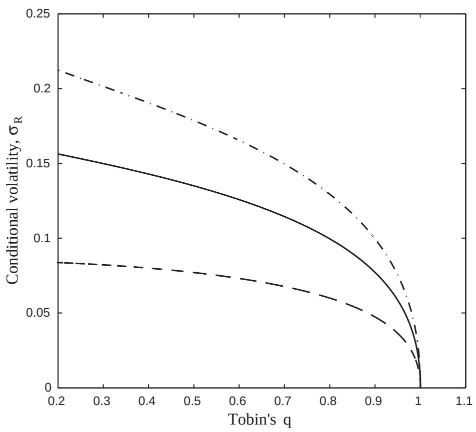

Figure 1: Conditional volatility of stock returns in the basic irreversible-investment model.

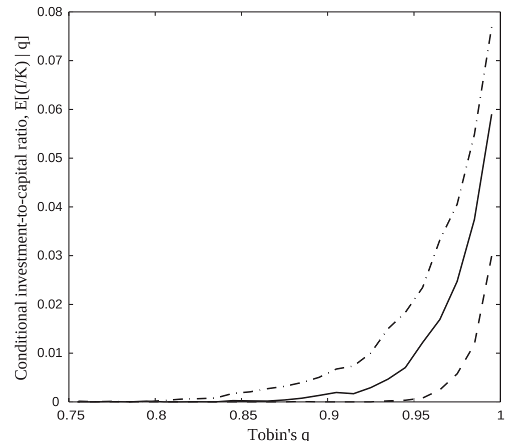

Figure 2: Conditional investment-to-capital ratio in the basic irreversible-investment model.

6.2 Figures 3 and 4: Bounded investment rate and nonmonotonicity

Read Figure 3:

- Figure 3 introduces the bounded investment-rate model.
- Investment remains low below $q = 1$, then increases sharply around $q = 1$.
- Because the investment rate is capped, q can exceed one.
- High q states represent constrained expansion rather than immediate adjustment.

Read Figure 4:

- Figure 4 plots conditional volatility against q when the investment rate is bounded.
- Volatility is high for low q, falls near $q = 1$, and rises again for high q.
- This is the paper's main nonmonotonic prediction.
- Low-q volatility comes from irreversibility; high-q volatility comes from the upper bound on investment.

Economic message: Adjustment costs create two different constrained regions. Both low q and high q can be high-volatility states, but for different real-side reasons.

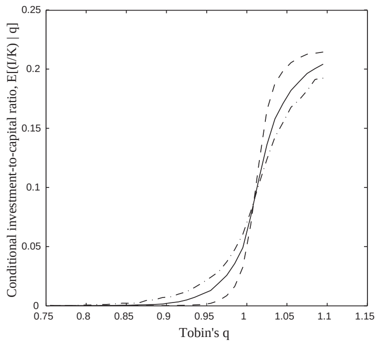

Figure 3: Conditional investment-to-capital ratio with a bounded investment rate.

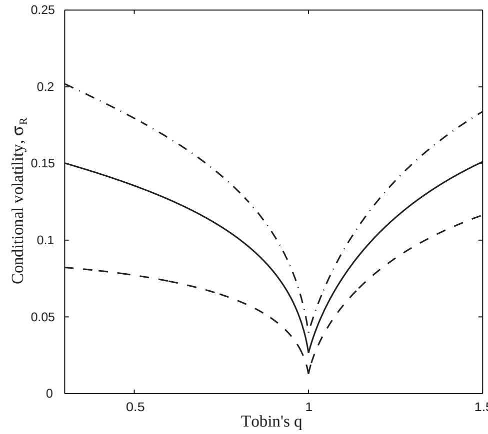

Figure 4: Conditional volatility with a bounded investment rate.

6.3 Tables 1 and 2: Conditional volatility regressions

Read Table 1:

- Table 1 estimates the quadratic volatility specification across 13 industry portfolios.
- Every point estimate of $a_{i,2}$ is positive.
- The average $a_{i,2}$ is 2.34 and statistically significant.
- The joint chi-square test rejects the null that all quadratic coefficients are zero.

Read Table 2:

- Table 2 estimates the piece-wise linear volatility specification.
- All low-side slope estimates $a_{i,1}$ are negative.
- The average low-side slope is -2.00 and statistically significant.
- The average high-side slope is 1.19 and statistically significant.
- The average difference between the high-side and low-side slopes is 3.19 and statistically significant.

Economic message: The volatility tests support the central prediction: conditional volatility is negatively related to market-to-book at low market-to-book and positively related to market-to-book at high market-to-book.

Table 1

Conditional volatility, quadratic specification

Estimates of the coefficients of the model $|R_{i,t} - \bar{R}_i| = a_{i,0} + a_{i,1}(M/B)_{i,t-1} + a_{i,2}(M/B)^2_{i,t-1} + \varepsilon_{i,t}$ based on the May 1963–December 2001 sample period and White's standard errors. $R_{i,t}$ denotes the excess monthly return on the industry portfolio i (in percent) and $(M/B)_{i,t-1}$ is the natural log of the market-to-book ratio of the portfolio at the end of the previous month, measured as a deviation from its mean. $\chi^2 = \hat{a}_2^\top \Sigma^{-1} \hat{a}_2$, where Σ is the White's estimate of the covariance matrix of $\hat{a}_2$. Under the null hypothesis that $a_2 = 0$, this statistic is asymptotically distributed as χ^2 with 13 degrees of freedom.

Portfolio	$a_{i,1}$	(Std. err.)	$a_{i,2}$	(Std. err.)
Nat. resources	−0.27	(0.73)	3.17	(2.22)
Construction	−1.14★	(0.48)	2.91★	(1.34)
Food, tobacco	−0.00	(0.39)	1.63★	(0.82)
Consumer products	−0.78	(0.57)	2.26	(1.56)
Logging, paper	−0.36	(0.31)	0.85	(0.49)
Chemicals	−1.29★	(0.45)	2.02★	(0.94)
Petroleum	0.36	(0.46)	2.77★	(0.82)
Mach., equipment	−0.40	(0.47)	4.41★	(0.96)
Transportation	−1.20★	(0.46)	0.34	(1.28)
Utilities, telecom.	0.02	(0.33)	1.91★	(0.74)
Trade	−0.67	(0.46)	1.32	(1.14)
Financial	−1.29★	(0.54)	4.46★	(1.55)
Services, other	0.69	(0.45)	2.42★	(0.95)
Average			2.34★	
(Std.err.)			(0.65)	
χ^2			37.98★	
(p-value)			(0.0003)	

Estimates and test statistics significant at 5% level are marked by a star.

Table 1: Conditional volatility, quadratic specification.

Table 2

Conditional volatility, piece-wise linear specification

Estimates of the coefficients of the model $|R_{i,t} - \bar{R}_i| = a_{i,0} + a_{i,1}(M/B)^-_{i,t-1} + a_{i,2}(M/B)^+_{i,t-1} + \varepsilon_{i,t}$ based on the May 1963–December 2001 sample period and White's standard errors. $R_{i,t}$ denotes the excess monthly return on the industry portfolio i (in percent) and $(M/B)_{i,t-1}$ is the natural log of the market-to-book ratio of the portfolio at the end of the previous month, measured as a deviation from its mean. $\chi^2 = \hat{c}^\top \Sigma^{-1} \hat{c}$, where c is the estimate of a coefficient and Σ is the White's estimate of the covariance matrix of c. Under the null hypothesis that $c = 0$, this statistic is asymptotically distributed as χ^2 with 13 degrees of freedom.

Portfolio	$a_{i,1}$	(Std. err.)	$a_{i,2}$	(Std. err.)
Nat. resources	−2.10	(1.41)	1.37	(1.46)
Construction	−2.67★	(1.05)	0.33	(0.95)
Food, tobacco	−1.31	(0.94)	1.46★	(0.60)
Consumer products	−2.21★	(0.98)	−0.05	(1.29)
Logging, paper	−1.07	(0.60)	0.54	(0.46)
Chemicals	−2.35★	(0.99)	0.19	(0.66)
Petroleum	−2.39★	(1.11)	3.67★	(0.95)
Mach., equipment	−3.85★	(1.14)	3.92★	(1.01)
Transportation	−1.48	(1.05)	−0.94	(0.86)
Utilities, telecom.	−0.79	(0.72)	1.07	(0.68)
Trade	−1.42	(0.95)	−0.08	(0.93)
Financial	−3.49★	(1.23)	1.33	(0.96)
Services, other	−0.91	(0.94)	2.60★	(1.00)
Average	−2.00★		1.19★	
(Std.err.)	(0.68)		(0.44)	
χ^2	29.33★		31.80★	
(p-value)	(0.0059)		(0.0026)	
Average ($a_{i,2} - a_{i,1}$)	3.19★			
(Std.err.)	(0.98)			

Estimates and test statistics significant at 5% level are marked by a star.

Table 2: Conditional volatility, piece-wise linear specification.

6.4 Table 3: Conditional expected returns

Read:

- Table 3 estimates the quadratic expected-return specification.
- The signs of the quadratic coefficients vary across industries.
- The average quadratic coefficient is 0.35, with a standard error of 1.15.
- The joint chi-square statistic has a p-value of 0.4393, so the null of no quadratic expected-return relation is not rejected.
- The expected-return results do not support the model's predicted pattern.

Economic message: The paper's empirical support is concentrated in conditional volatility, not expected returns. This is consistent with the author's view that expected-return implications are more fragile and harder to test.

Table 3
Conditional expected return, quadratic specification

Estimates of the coefficients of the model $R_{i,t} = a_{i,0} + a_{i,1}(M/B)_{i,t-1} + a_{i,2}(M/B)^2_{i,t-1} + \varepsilon_{i,t}$ based on the May 1963–December 2001 sample period and White's standard errors. $R_{i,t}$ denotes the excess monthly return on the industry portfolio i (in percent) and $(M/B)_{i,t-1}$ is the natural log of the market-to-book ratio of the portfolio at the end of the previous month, measured as a deviation from its mean. $\chi^2 = \hat{a}_2^\top \Sigma^{-1} \hat{a}_2$, where Σ is an estimate of the covariance matrix of $\hat{a}_2$, robust with respect to conditional heteroskedasticity. Under the null hypothesis that $a_2 = 0$, this statistic is asymptotically distributed as χ^2 with 13 degrees of freedom.

Portfolio	$a_{i,2}$	(Std. err.)
Nat. resources	−1.43★	(3.48)
Construction	0.61★	(2.19)
Food, tobacco	−0.96	(1.19)
Consumer products	−0.05★	(2.39)
Logging, paper	−0.14	(0.79)
Chemicals	1.35	(1.47)
Petroleum	0.36	(1.45)
Mach., equipment	−0.35	(1.79)
Transportation	2.90★	(2.06)
Utilities, telecom.	0.89	(1.25)
Trade	−0.24	(1.73)
Financial	1.66★	(2.34)
Services, other	−0.07	(1.60)
Average	0.35	
(Std.err.)	(1.15)	
χ^2	13.11	
(p-value)	(0.4393)	

Estimates and test statistics significant at 5% level are marked by a star.

Table 3: Conditional expected return, quadratic specification.

7 Short summary

- The paper studies how real investment frictions affect stock-return dynamics.
- The model is a two-sector production economy with one investment-constrained sector.
- In the basic model, investment into the constrained sector is irreversible.
- When q is low, firms cannot disinvest, so capital supply is inelastic and stock prices absorb shocks.
- This implies high conditional volatility at low q and lower volatility as q approaches one.
- In the full model, the investment rate is bounded above.
- With a bounded investment rate, q can exceed one because firms cannot expand instantaneously.
- High q also becomes a constrained state, so volatility rises again at high q.
- The main theoretical prediction is a nonmonotonic relation between market-to-book and conditional volatility.
- The empirical analysis uses 13 U.S. industry portfolios from May 1963 to December 2001.
- Market-to-book is measured using the log of industry market-to-book, demeaned by industry.
- Conditional volatility is proxied by the absolute deviation of monthly excess returns from the industry mean.
- The quadratic volatility regressions show positive convexity in all 13 industries.

- The piece-wise volatility regressions show negative slopes at low market-to-book and positive slopes at high market-to-book.
- The expected-return regressions do not provide statistically significant support.
- The key takeaway is that market-to-book predicts return volatility because it summarizes the state of real investment constraints.

8 Conclusion

8.1 Core conclusion

Kogan shows that real investment frictions can generate state-dependent stock-return volatility. When investment is irreversible or when the investment rate is bounded, capital supply becomes inelastic in some states. In those states, prices must absorb demand shocks, leading to higher return volatility.

8.2 Economic intuition

The model's intuition is a price-versus-quantity adjustment story. If firms can adjust capital, shocks are partly absorbed by investment quantities. If firms cannot adjust capital, shocks are absorbed by prices. Low market-to-book identifies states in which firms would like to shrink but cannot. High market-to-book identifies states in which firms would like to expand faster than adjustment costs allow. Both states can produce high volatility.

8.3 Incremental contribution revisited

The paper provides a real-side explanation for why firm characteristics such as market-to-book forecast return dynamics. Market-to-book matters not only because it contains price, but because it reveals the investment state of the industry. This connects production-based asset pricing to empirical return predictability.

8.4 Implications

The paper implies that empirical asset-pricing models should allow nonlinear relations between characteristics and conditional moments. A single linear market-to-book slope can miss the fact that low and high market-to-book states are both constrained states, while intermediate market-to-book states are less constrained.

8.5 Limitations

The empirical analysis supports the volatility predictions but not the expected-return predictions. The expected-return tests are difficult because expected returns are noisy and the model's expected-return implications rely on additional assumptions about systematic risk. The empirical proxy for q is also imperfect, and the industry-level regressions are reduced-form tests rather than a full structural estimation.

8.6 Takeaway

The enduring takeaway is that asset prices reflect real investment constraints. Market-to-book ratios help forecast return volatility because they indicate whether capital supply is elastic or constrained. Real investment irreversibility and adjustment costs therefore provide a concrete production-based mechanism linking firm characteristics to stock-return dynamics.